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Sciences Appliquées

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Synchronous generators

ELEC0447 - Analysis of electric power and energy systems

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Overview

1. Operating principle

2. Types of synchronous machines

3. Equivalent circuit and modeling

4. Properties and reactive power control

5. Generators in power flow analysis

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What will we learn today?

- What synchronous generators are and how they work.
- How we can model them for power system analyses.
- How to include generator capability limits in power flow computations.

You will be able to solve exercises 9.5 to 9.9 of Ned Mohan's book:

Mohan, Ned. *Electric power systems: a first course*. John Wiley & Sons, 2012.

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Operating principle

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Synchronous machines

Synchronous machines produce the major part of electrical energy worldwide, with unit ratings ranging from a few kVA to over 1500 MVA (nuclear power plants).

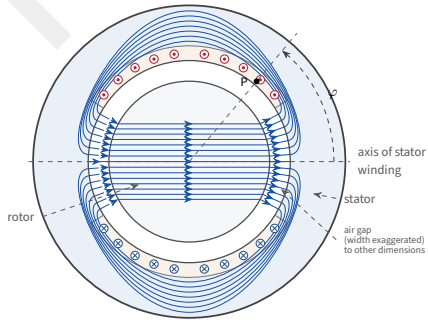
Beyond bulk active power production, they play three critical system roles:

- **Frequency setting**: They impose the synchronous frequency of grid voltages and currents.
- **Energy buffer**: They store significant kinetic energy in their rotating masses (providing instantaneous system inertia).
- **Voltage control**: They can produce or consume reactive power by adjusting their rotor field current.

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Magnetic field created by the stator

- **Stator (armature)**: Stationary outer structure separated from the rotor by a small air gap.
- Subjected to time-varying magnetic flux $\Longrightarrow$ built from thin steel laminations to minimize **eddy (Foucault) currents**.
- Equipped with three windings (a, b, c) distributed 120° apart in space.
- Magnetic field lines cross the air gap in a radial direction.



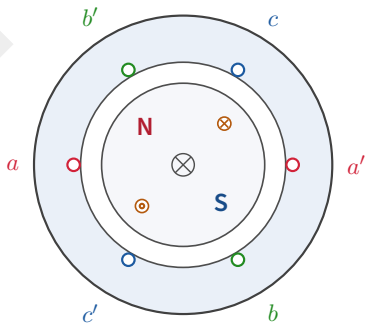
Field from direct current in one stator winding

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Three-phase stator winding layout

The amplitude $B(\varphi)$ of the flux density along the air gap:

- Is a periodic function of spatial angle φ with period 2π.
- Has a discrete "staircase" shape in practice due to discrete slots.
- Is shaped as close as possible to a pure sinusoid by carefully distributing conductor coils along stator slots.



Three-phase winding layout (single turn shown)

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Rotating magnetic field I

Total magnetic flux density created by the three phases at spatial angle φ:

$$B_{3\phi}(\varphi) = ki_a \cos\varphi + ki_b \cos\left(\varphi - \frac{2\pi}{3}\right) + ki_c \cos\left(\varphi - \frac{4\pi}{3}\right)$$

When balanced three-phase sinusoidal currents flow in the windings:

$$i_a(t) = \sqrt{2}I\cos(\omega t + \psi)$$
$$i_b(t) = \sqrt{2}I\cos\left(\omega t + \psi - \frac{2\pi}{3}\right), \quad i_c(t) = \sqrt{2}I\cos\left(\omega t + \psi - \frac{4\pi}{3}\right)$$

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Rotating magnetic field II

Using trigonometric identities:

$$B_{3\phi}(\varphi, t) = \frac{3\sqrt{2}kI}{2} \cos(\omega t + \psi - \varphi)$$

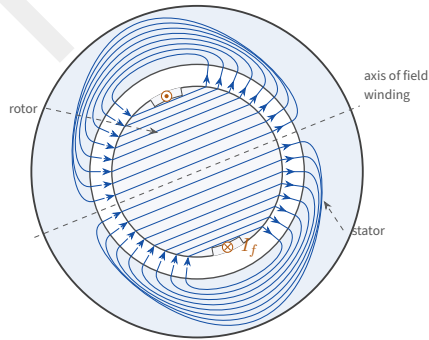
Key conclusion

The three-phase AC currents together produce a magnetic field of constant amplitude rotating at electrical angular speed $\omega = 2\pi f$ (Ferraris theorem).

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Magnetic field created by the rotor

- **Rotor**: The rotating cylindrical or salient part mounted on the turbine shaft.
- In steady state, a direct current (I_f) flows through the rotor winding.
- This winding is known as the **field winding** (*enroulement d'excitation*).
- The DC current creates a magnetic field with alternating North and South poles that rotates along with the rotor.

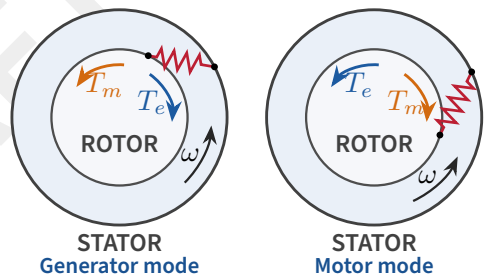


Magnetic field produced by rotor field winding

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Interaction between fields & electromechanical conversion

- In steady state, the rotor rotates at the exact same angular speed ω as the stator magnetic field: **synchronism**.
- Therefore, the stator and rotor magnetic fields are stationary with respect to each other.
- Both fields tend to align in the manner of two coupled magnets.
- Displacing them creates an electromagnetic torque T_e.
- There exists a maximum torque $T_{e,\max}$ beyond which synchronism is lost.



Mechanical analogy: generator mode ↔ motor mode

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Types of synchronous machines

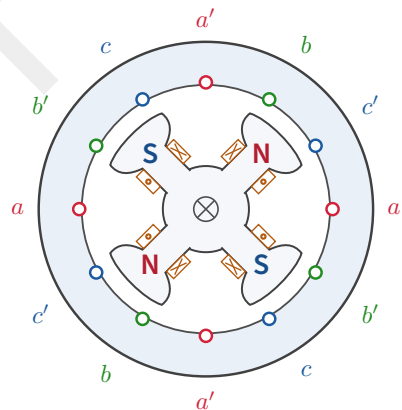
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Machines with multiple pole pairs

Certain prime movers (e.g. hydro turbines) run at low rotational speed, yet the electrical frequency must remain $f = 50$ Hz ($T = 20$ ms):

- The rotor is equipped with p pairs of poles ($2p$ poles).
- In one mechanical revolution, each stator winding experiences p electrical cycles.
- Rotational speed n (in rpm) is linked to grid frequency f:

$$n = \frac{60f}{p} = \frac{3000}{p} \text{ rpm} \quad (\text{at } 50 \text{ Hz})$$



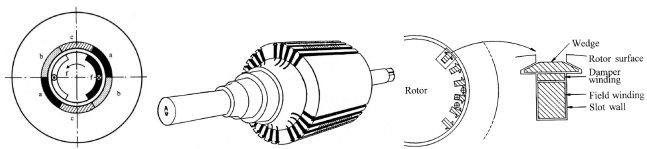
Example: $p = 2$ (4 poles, $n = 1500$ rpm)

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Round-rotor generators (turbo-alternators)

Driven by high-speed steam or gas turbines:

- Typically $p = 1$ (3000 rpm) or $p = 2$ (1500 rpm in nuclear plants).
- Cylindrical rotor forged from solid steel alloy; small diameter and long length (due to severe centrifugal forces).
- Field winding embedded in milled rotor slots.
- Extreme power density: cooled by pressurized hydrogen ($7\times$ higher thermal conductivity than air) or demineralized water in hollow conductors.



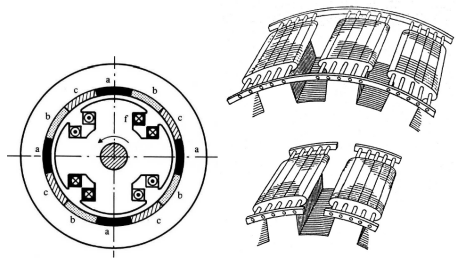
Cross-section of a round-rotor machine

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Salient-pole generators

Driven by low-speed hydraulic turbines or diesel engines:

- Number of pole pairs is large ($p \geq 4$, often $p \in [10, 40]$).
- Protruding (*salient*) poles with concentrated field coils mounted on a large rotor wheel.
- Large diameter, short axial length.
- Rotor poles are made of laminated steel sheets.
- Usually cooled by natural or forced airflow around the rotor.



Cross-section of a salient-pole machine

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Damper windings (*amortisseurs*)

- Conductive bars (copper/brass) placed in rotor pole shoes and short-circuited at ends with rings (similar to a squirrel cage).
- **In steady state**: The stator and rotor magnetic fields rotate synchronously $\Longrightarrow$ no relative motion, zero induced current in the dampers.
- **During a disturbance**: The rotor accelerates or decelerates relative to the synchronous magnetic field $\Longrightarrow$ currents are induced in the damper bars.
- According to Lenz's law, these induced currents produce a **damping torque** that counteracts rotor oscillations and helps restore synchronism.
- In solid round-rotor machines, eddy currents flowing directly in the rotor steel provide a similar intrinsic damping effect.

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Equivalent circuit and modeling

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Induced electromotive force (emf)

Two separate magnetic sources induce voltages in the stator windings:

1. **Excitation field**: The rotating rotor field induces an internal electromotive force (emf) denoted $\bar{E}_{af}$ (or open-circuit voltage).
2. **Armature reaction**: The AC currents flowing in the stator windings produce their own magnetic field, inducing a voltage drop modeled as $-jX_s\bar{I}_a$.

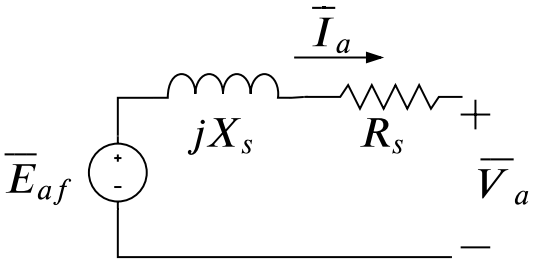
Assuming no magnetic saturation, the resulting terminal phase voltage $\bar{V}_a$ is obtained by superposition:

$$\bar{V}_a = \bar{E}_{af} - R_s\bar{I}_a - jX_s\bar{I}_a$$

where R_s is the stator winding resistance and X_s is the **synchronous reactance**.

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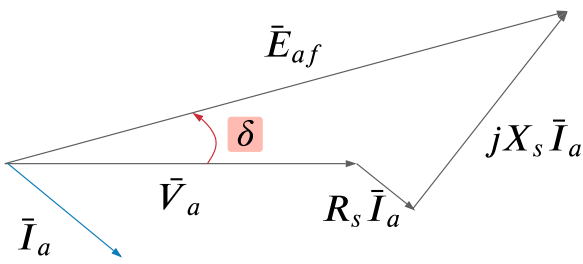
Per-phase equivalent circuit



$$\bar{V}_a = \bar{E}_{af} - (R_s + jX_s)\bar{I}_a$$

- R_s: Stator resistance.
- X_s: Synchronous reactance (leakage + magnetizing reactance).
- δ: Angle between $\bar{E}_{af}$ and $\bar{V}_a$, called the **internal angle** or **load angle**.

Phasor diagram



- $\bar{V}_a = V_a \angle 0°$ is chosen as the reference.
- $\bar{I}_a$ is the stator phase current with phase lag ϕ.
- Resistive drop $R_s\bar{I}_a$ is parallel to $\bar{I}_a$.
- Reactive drop $jX_s\bar{I}_a$ leads $\bar{I}_a$ by $90°$.
- Vector sum yields the internal emf:

$$\bar{E}_{af} = \bar{V}_a + R_s\bar{I}_a + jX_s\bar{I}_a$$

Parameters and transient reactances

Typical machine parameters (in per unit on machine rating base):

- Stator resistance: $R_s \approx 0.005$ pu (frequently neglected in grid studies).
- Synchronous reactance: $X_s \in [1.5, 2.5]$ pu (for round-rotor turbo-generators).

Machine models for transient conditions: Depending on the time frame of the phenomenon, stator reaction is hindered by rotor screening circuits:

- **Subtransient state** (first few cycles after a short circuit): $X_s'' \ll X_s$ $(0.15 - 0.25$ pu).
- **Transient state** (hundreds of milliseconds): $X_s' < X_s$ $(0.25 - 0.40$ pu).
- **Steady state**: X_s $(1.5 - 2.5$ pu).

$$\implies X_s'' < X_s' < X_s$$

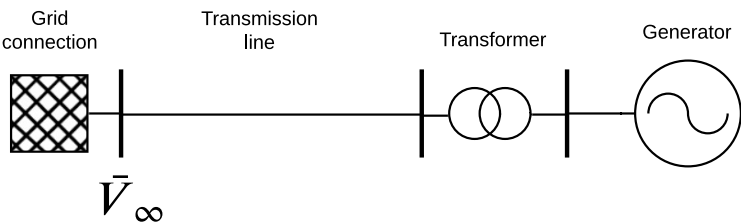
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Properties and reactive power control

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Power transfer to an external grid I

Consider a generator connected to an infinite bus ($\bar{V}_\infty = V_\infty \angle 0°$) through a step-up transformer and transmission line:



Power transfer to an external grid II

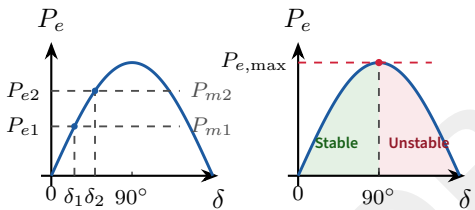
Neglecting resistances, let $X_T = X_s + X_{tr} + X_{line}$ be the total reactance between $\bar{E}_{af}$ and $\bar{V}_\infty$. The three-phase active power delivered to the grid is:

$$P = \frac{E_{af} V_\infty}{X_T} \sin \delta$$

where δ is the angle between internal emf $\bar{E}_{af}$ and infinite bus voltage $\bar{V}_\infty$.

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Steady-state stability limit



(a) Operating points **(b) Stability limit**

- When excitation E_{af} is constant, active power P is strictly proportional to $\sin\delta$.
- Maximum power is reached at $\delta = 90°$:

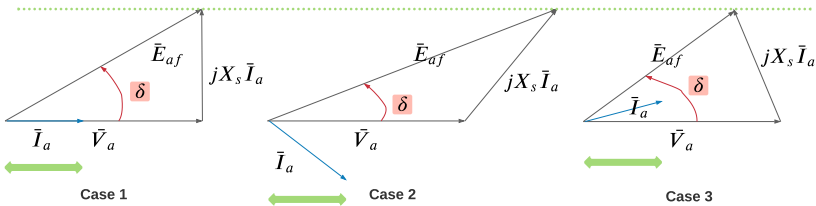
$$P_{\max} = \frac{E_{af} V_\infty}{X_T}$$

- For $\delta > 90°$, $\frac{dP}{d\delta} < 0$: additional mechanical power causes rotor acceleration, increasing δ and reducing electrical power $\Longrightarrow$ **loss of synchronism**.
- In operation, δ is kept well below $90°$, typically $\delta \leq 40°$.

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Adjusting reactive power via field current

Suppose the generator delivers constant active power P, requiring $E_{af} \sin \delta =$ cst and $I_a \cos \phi =$ cst:



By varying the DC field excitation current I_f, we directly control the amplitude E_{af} and the reactive power Q exchanged with the grid.

Overexcitation vs. Underexcitation

Overexcitation ($I_f \nearrow$, $E_{af} \nearrow$):

- Current $\bar{I}_a$ lags terminal voltage $\bar{V}_a$ ($\phi > 0$).
- Generator **supplies** reactive power to the grid:

$$Q = 3V_a I_a \sin\phi > 0$$

- The generator behaves as a **capacitor** connected to the power system.

Underexcitation ($I_f \searrow$, $E_{af} \searrow$):

- Current $\bar{I}_a$ leads terminal voltage $\bar{V}_a$ ($\phi < 0$).
- Generator **absorbs** reactive power from the grid:

$$Q = 3V_a I_a \sin\phi < 0$$

- The generator behaves as an **inductor** connected to the power system.

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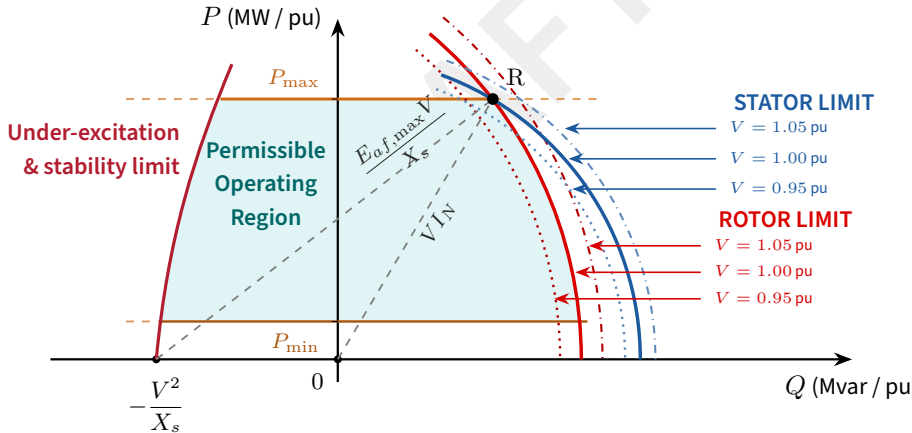
Voltage regulation and synchronous condensers

- **Automatic Voltage Regulation (AVR)**:
  - Controls the DC rotor excitation current I_f via feedback control.
  - Dynamically adjusts internal emf E_{af} and reactive power Q to regulate the terminal bus voltage to a target setpoint V_0.

- **Synchronous Condensers**:
  - Synchronous machines running without a mechanical load (drawing only enough active power to cover internal losses).
  - Used exclusively to supply or absorb continuously variable reactive power and provide short-circuit power and inertia to the grid.

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Generator capability curve: the permissible operating region in the (P, Q) plane under constant terminal voltage V



Operating limits breakdown

1. **Prime mover (turbine) limit**: Maximum mechanical power input ($P \leq P_{\max}$). A minimum power $P_{\min}$ also exists for flame stability in thermal boilers.
2. **Stator thermal limit**: Armature Joule losses ($R_s I_a^2$). Current magnitude is capped ($I_a \leq I_{a,\max}$):
$$P^2 + Q^2 \leq (V I_{a,\max})^2 \quad \text{(circle centered at origin)}$$
3. **Rotor thermal limit**: Joule losses in field winding ($R_f I_f^2$). Maximum excitation voltage $E_{af} \leq E_{af,\max}$:
$$P^2 + \left(Q + \frac{V^2}{X_s}\right)^2 \leq \left(\frac{V E_{af,\max}}{X_s}\right)^2 \quad \text{(circle centered at } (-V^2/X_s, 0))$$
4. **Underexcitation / Stability limit**: Minimum field excitation required to avoid excessive internal angle δ and loss of synchronism.

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Generators in power flow analysis

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Inclusion in power flow: PV to PQ bus switching

- **Initial PV bus representation**:
  - Active power P is fixed by turbine dispatch.
  - Voltage magnitude $|V|$ is fixed by the AVR setpoint.
  - Reactive power Q and phase angle θ are state variables to solve.
- **Limit verification**: Computed Q is checked against capability limits: $Q_{\min} \leq Q \leq Q_{\max}$.
- **PV to PQ switching**: If a limit is violated (e.g. $Q > Q_{\max}$):
  - Generator field current cannot sustain the voltage setpoint.
  - Bus is switched from a **PV bus** to a **PQ bus** with $Q = Q_{\max}$ (or $Q_{\min}$).
  - Voltage magnitude $|V|$ is freed and becomes a state variable.
  - Power flow is re-solved iteratively until no limits are violated.

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Simulation with PandaPower

In PandaPower, generator limits can be directly enforced during AC power flow calculations:

```
pp.runpp(net, enforce_q_limits=True)
```

Google Colab Notebook

Explore synchronous machine modeling and PV/PQ switching in PandaPower:
https://colab.research.google.com/drive/1Q9E62wzrKiCsBIyRH9JnCTMSibSlY9OG?usp=sharing

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References

- Course notes of ELEC0014 (*Electric Energy Systems*) by Pr. Thierry Van Cutsem, University of Liège.
- Mohan, Ned. *Electric power systems: a first course*. John Wiley & Sons, 2012.
- L. Thurner, A. Scheidler, F. Schäfer et al., "pandapower - an Open Source Python Tool for Convenient Modeling, Analysis and Optimization of Electric Power Systems", *IEEE Transactions on Power Systems*, vol. 33, no. 6, pp. 6510-6521, Nov. 2018.

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